

Extended Class A and Simul-Class TM Technology



Ampbooks

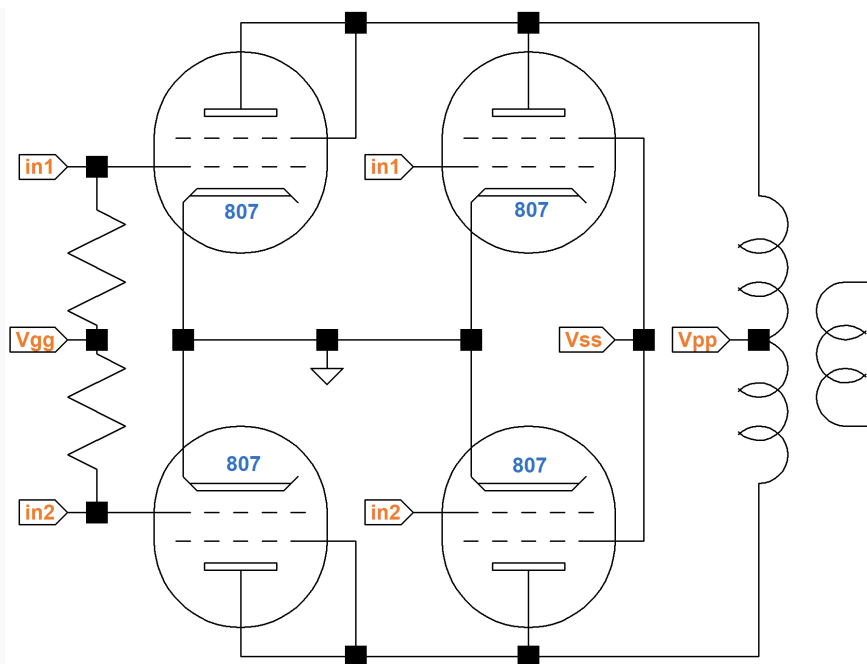
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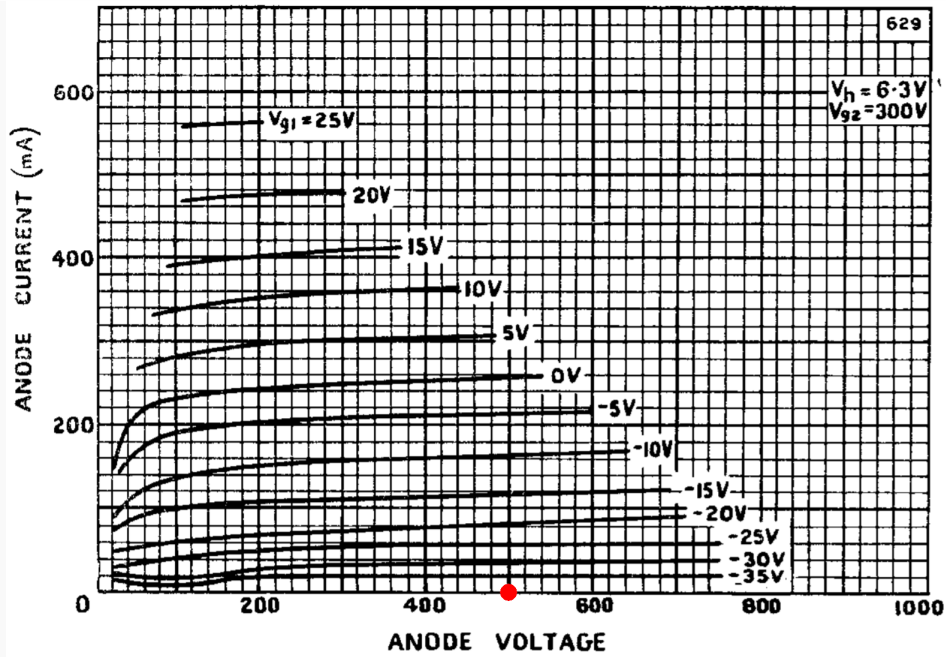
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Extended Class A

Extended Class A is a technique that uses a triode and a pentode (or beam power tetrode) on both sides of a push-pull power amp.¹ Here is an implementation described by Howard Sterling in 1951.²

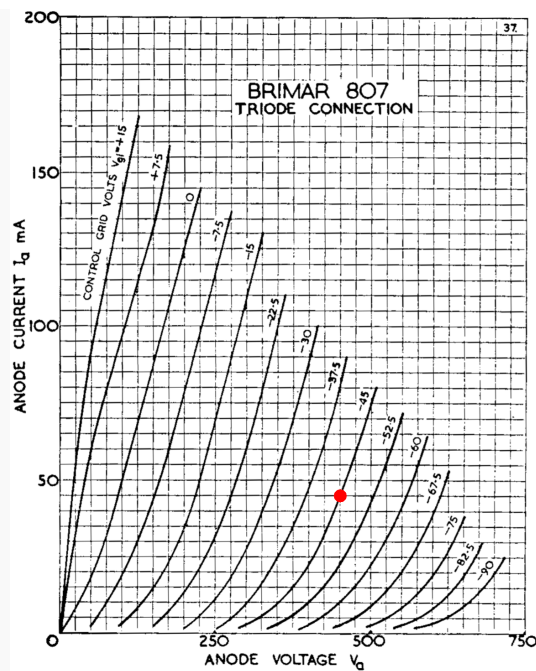


There are four 807 beam power tetrodes in the circuit. On each side of the transformer, one tube is triode-connected and the other is tetrode-connected.³ The supply voltages are $V_{PP} = 450V$, $V_{SS} = 300V$, and $V_{GG} = -45V$. The plate characteristics for a tetrode-connected 807 with a 300V screen show that the two tetrodes are in cutoff when there is no signal.



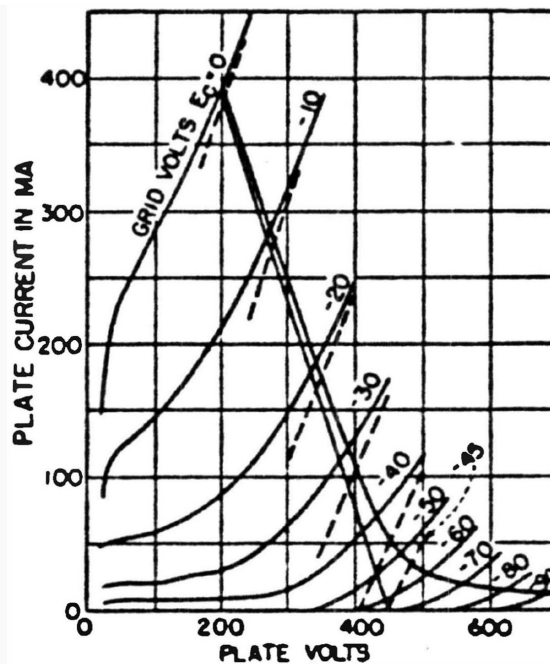
The guitar signal needs to cause the grid voltage to increase to about $-40V$ for conduction to occur. Otherwise the tetrodes are essentially disconnected from the circuit.

For a 450V plate supply, the idle screen voltage for the triode-connected tubes is 150V greater: 450V. According to the triode-connected curves, this causes 45mA of plate current to flow.



The tetrodes are in cutoff at idle or when the amp is operating at a low power level. They are effectively disconnected from the audio circuit and the amp runs pure Class A with only the triodes in operation. At high power levels the tetrode-connected tubes kick in for an efficient power boost.

Here are composite plate characteristics² for 807s operating in extended Class A with $V_{PP} = 450V$, $V_{SS} = 300V$, and $V_{GG} = -45V$.



The load line is for a plate-to-plate transformer primary impedance of $2.5k\Omega$. The dashed curves are the composite grid lines for push-pull operation. When the AC guitar signal is 0V, for example, both grids are at $-45V$. If the plates are at $450V$, the tubes draw equal plate current in opposite directions for a net plate current of zero.

*"The first and most important point of interest is the path of operation for one side. Even with maximum grid swing, operation is Class A; in fact, the path of operation is nowhere near the zero axis. In effect, the triode characteristics are simply elevated by the current drawn by the tetrode, and throughout the region traversed by the load line the performance is typical of a triode."*²

The plate voltage swings from $450V$ to $200V$, i.e. $250V$ peak. Plate current is $400mA$ peak. Full power is therefore approximately

$$(250V)(400mA) / 2 = 50W$$

According to the RCA 807 data sheet, with the screens at 300V, two tetrode-connected tubes in conventional Class AB push-pull deliver a maximum of 36W for a 400V plate supply and a -30V bias. For a 500V plate supply and a -32V bias, full power is 46W. For four tubes the power doubles to 72W and 92W, respectively, so there is a power penalty for extended Class A compared to a more traditional Class AB design.

The tetrodes are in cutoff at idle or when the amp is operating at a low power level, so they can be cut out of the circuit entirely without affecting power amp operation. This gave Sterling the following idea to add to his patent application.

"This suggests another variation: here is the convenient place for a high-level, low-level switch ... As soon as patent arrangements, now in progress, are completed, a commercial version of this circuit will be made available." 2

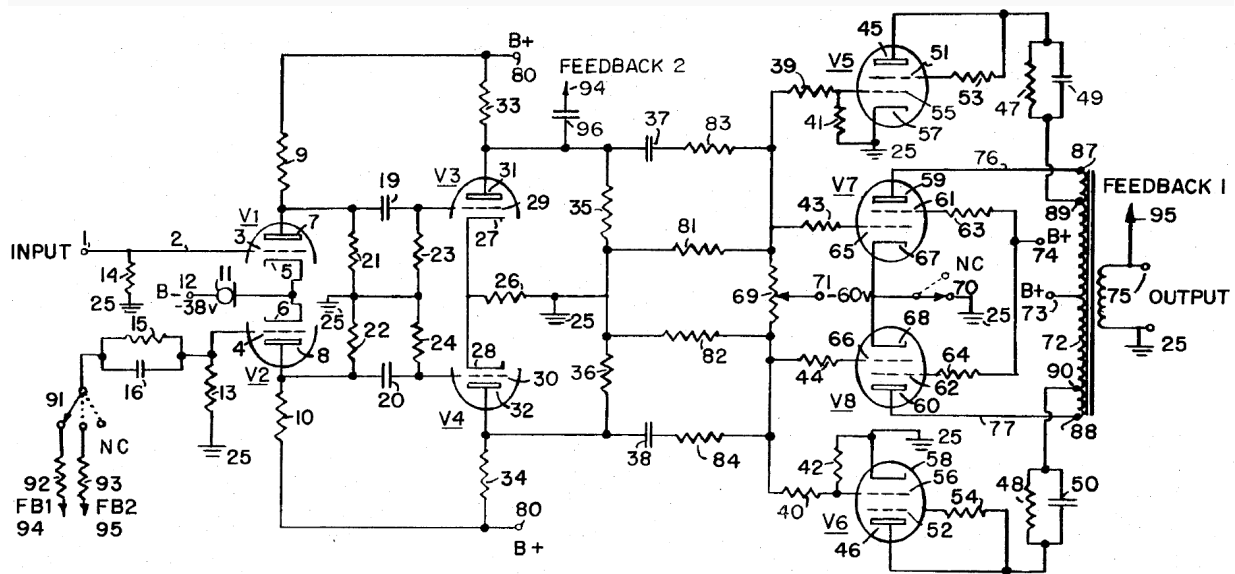
Apparently, a patent was never awarded.

Mesa/Boogie Simul-Class™

In the early 1980s, Randall Smith, who was familiar with the concept of extended Class A as built by Lloyd Hust,⁴ patented a technology called *Simul-Class.™* One of his goals was to eliminate the crossover distortion that can occur with traditional Class AB while still retaining its advantages.

"Although the [Class AB] output devices operate Class A at low power, they become more and more Class B when driven harder and a somewhat harsh sounding distortion with an abrupt onset and visible crossover occurs at the crucial time: at clip. The power output and efficiency with Pentodes in an AB arrangement is fairly high, however." 5

Here is an embodiment from one of the patent documents.

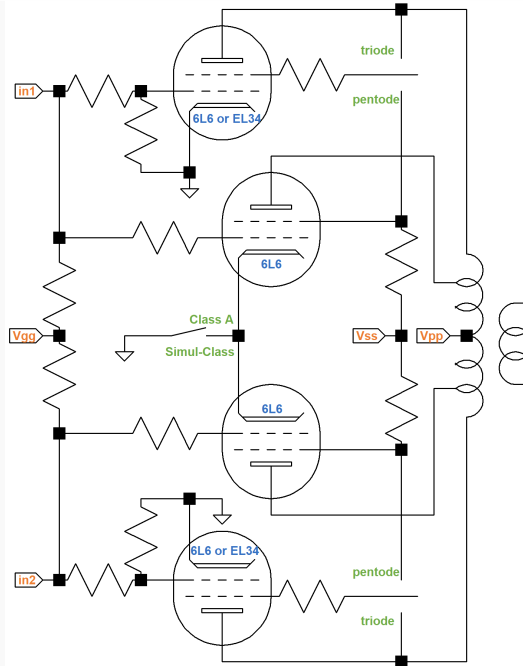


V7 and V8 are tetrode-connected and biased for Class AB or Class B. They can be switched into the circuit via switch 70 when additional output power is desired. Unlike Sterling's circuit, the effective DC grid bias is more negative for the tetrodes (or pentodes) than for the triodes. This is accomplished through the voltage dividers formed by resistors 39 and 41 and by resistors 40 and 42. The transition from 2-tube to 4-tube operation therefore occurs at a higher signal level.

Triodes V1 and V2 form a long-tailed-pair phase inverter with "constant current source device" 11 at the tail, which provides "excellent linearity and accurate phase inversion." V3 and V4 form a differential voltage amplifier that is designed to clip slightly before the power tubes begin clipping, making the transition to overdrive "soft and gradual."

Global negative feedback from the output transformer secondary to the LTP phase inverter (Feedback 1) or local feedback (Feedback 2) can be switched into the circuit via switch 91.

The Mesa/Boogie Mark IV uses a traditional LTP phase inverter in front of the power tubes, eliminating the constant current source device at the tail and the intervening differential amplifier. The inner 6L6 tubes are tetrode-connected and can be disconnected from the circuit via a switch between the cathodes and ground. The outer power tubes, which are 6L6 tetrodes or EL34 pentodes, can be switched between triode and tetrode (pentode) operation.



The power amp design was highly successful and the Mark IV was in steady production for over 19 years. According to the Mesa/Boogie web site, there are 50 thousand Mark II, III, and IVs still in active service.

"All this and the elusive magic of Simul-Class™ ...our patented way of enriching a power section. Think of it as two different power amps working simul-taneously. One extracts the juice of Class A sweetness while the other delivers the high power punch of Pentode Class AB." 6

References

1F. Langford-Smith, ed., **Radiotron Designer's Handbook**, 4th ed., (Harrison: RCA, 1953), p. 587.

2Howard T. Sterling, "Extended Class-A Audio," **Electronics**, May 1951, pp. 101-103.

3Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018), p. 106.

4Lloyd B Hust, "Extended Class A Amplifier," **Radio and Television News**, September 1953, pp. 40-42, 146-148.

5U.S. Patents [4,532,476](#) and [4,593,251](#), Randall C. Smith, inventor.

6Available at <https://mesaboogie.com/support/out-of-production/mark-iv.html> (Retrieved April 12, 2020)